

Desenvolvimento e Análise de Modelos Clássicos de Recuperação de Informação

STUDENT

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Contents

Introdução	2
Técnicas utilizadas	2
Tratamento de Documentos/Queries	2
Posting Lists	3
Modelo Probabilístico BM25	3
Modelo Vetorial	4
Avaliação dos modelos	4
Comparação entre Modelo Vetorial e Modelo Probabilístico	4
Curva Precision x Recall	4
Comparação de MAP	7
Análise por consulta	7
Consultas com BM25 superior	7
Consultas com Modelo Vetorial superior	10
Consultas com desempenho insatisfatório em ambos	16
Variação de parâmetros do BM25	20
MAP agrupado por parâmetros	20
Mapa de bolhas P@10 x R@10	22

Introdução

Este trabalho consiste na implementação e análise de um sistema de Recuperação Textual utilizando dois modelos clássicos de recuperação da informação: modelo vetorial e modelo probabilístico. Ao analisar e comparar os dois modelos, é esperado obter uma distinção clara de classificação diante de uma base de dados real, fazendo uso de métricas exclusivas para avaliação de sistemas de Recuperação da Informação.

O sistema consiste basicamente em 4 etapas:

1. Processamento de documentos
2. Criação de posting lists
3. Processamento de query
4. Aplicação de query em modelos de RI

Todas foram implementadas na linguagem Python.

Técnicas utilizadas

Tratamento de Documentos/Queries

Para um processamento adequado dos documentos e queries, foi escolhido o uso da biblioteca NLTK (Natural Language Toolkit) do Python, que possui ferramentas úteis para realizar o processamento, entre elas:

- Lista de stopwords
- Lematização

- Stemming

As funções de tokenização e normalização foram aplicadas em hard-code.

Também foram criadas 4 possibilidades de pipeline:

- **NoStopRemovalNoStemming** - Sem remoção de stop words, sem stemming
- **NoStopRemovalWithStemming** - Sem remoção stop words, com stemming
- **WithStopRemovalNoStemming** - Com remoção de stop words, sem stemming
- **WithStopRemovalWithStemming** - Com remoção de stop words, com stemming

Posting Lists

A posting list (ou índice invertido) é uma estrutura de dados que mapeia uma palavra específica para uma lista de documentos onde isto aparece, adicionado de informações opcionais.

Na implementação deste trabalho, a posting list é representada por uma classe que possui quatro mapeamentos:

1. **Postings** - Dicionário que mapeia termos para um dicionário de IDs que mapeia para a frequência do termo no documento
2. **Term Frequencies** - Dicionário que mapeia termos para a frequência total na posting list (`tf`)
3. **Document Frequencies** - Dicionário que mapeia termos para o número de documentos em que aparecem (`df`)
4. **Document Lengths** - Dicionário que mapeia IDs para o número total de termos no documento

As funções públicas da classe são para resgatar informações concernentes a estes quatro dicionários.

Modelo Probabilístico BM25

O modelo probabilístico BM25 consiste na seguinte função de ranqueamento:

$$\text{score}(d, q) = \sum_{t \in q} IDF(t) \frac{tf(t, d)(k_1 + 1)}{(tf(t, d) + k_1(1 - b + \frac{b(|d|)}{\text{avgdl}}))}$$

- $tf(t, d)$ - Frequência do termo no documento
- k_1 - Parâmetro de controle da saturação
- b - Parâmetro de controle da influência do tamanho do documento
- $|d|$ - Tamanho do documento

Onde:

$$IDF(t) = \log \left(1 + \frac{N - n_t + 0.5}{n_t + 0.5} \right)$$

$$\text{avgdl} = \frac{1}{N} \sum_i^N |d_i|$$

- N = Quantidade total de documentos
- n_t = Quantidade de documentos que contém o termo t
- $|d_i|$ = Tamanho do documento

A implementação desta função foi feita exatamente como descrito aqui, com uso das bibliotecas `math` e `numpy`. Sua implementação foi feita como classe, recebendo os seguintes parâmetros:

- Posting List - Instância da classe `PostingList`
- Pipeline Options - Seleção de modo de processamento das queries

Modelo Vetorial

O modelo vetorial consiste na seguinte função de ranqueamento:

$$\text{sim}(d, q) = \frac{\sum w_{dt} w_{qt}}{\sqrt{\sum w_{dt}^2} + \sqrt{\sum w_{qt}^2}}$$

Onde:

$$w = \begin{cases} (1 + \log(f(t))) \cdot \log\left(\frac{N}{n_t}\right), & \text{se } f(t) > 0 \\ 0, & \text{se } f(t) = 0 \end{cases}$$

- $f(t)$ - Frequência do termo no documento/query.

Para otimizar o tempo de execução do algoritmo, o vetor de pesos é montado utilizando somente os termos pertencentes à query, com valor igual a zero caso algum dos termos seja inexistente no documento. Assim, se existem `n` termos na query, o vetor de qualquer documento selecionado terá dimensionalidade `n`, assim como o vetor de pesos da query.

A implementação desta função foi feita exatamente como descrito aqui, com uso das bibliotecas `math` e `numpy`. Sua implementação foi feita como classe, recebendo os seguintes parâmetros:

- Posting List - Instância da classe `PostingList`
- Pipeline Options - Seleção de modo de processamento das queries

Avaliação dos modelos

Ambos os modelos foram avaliados sob as seguintes condições:

- Para ambos, foram testados os 4 tipos de pipeline
- Para o modelo probabilístico, foram testadas as 9 combinações possíveis para os parâmetros k_1 e b

$$k_1 \in \{0.5, 1.2, 2.0\}$$

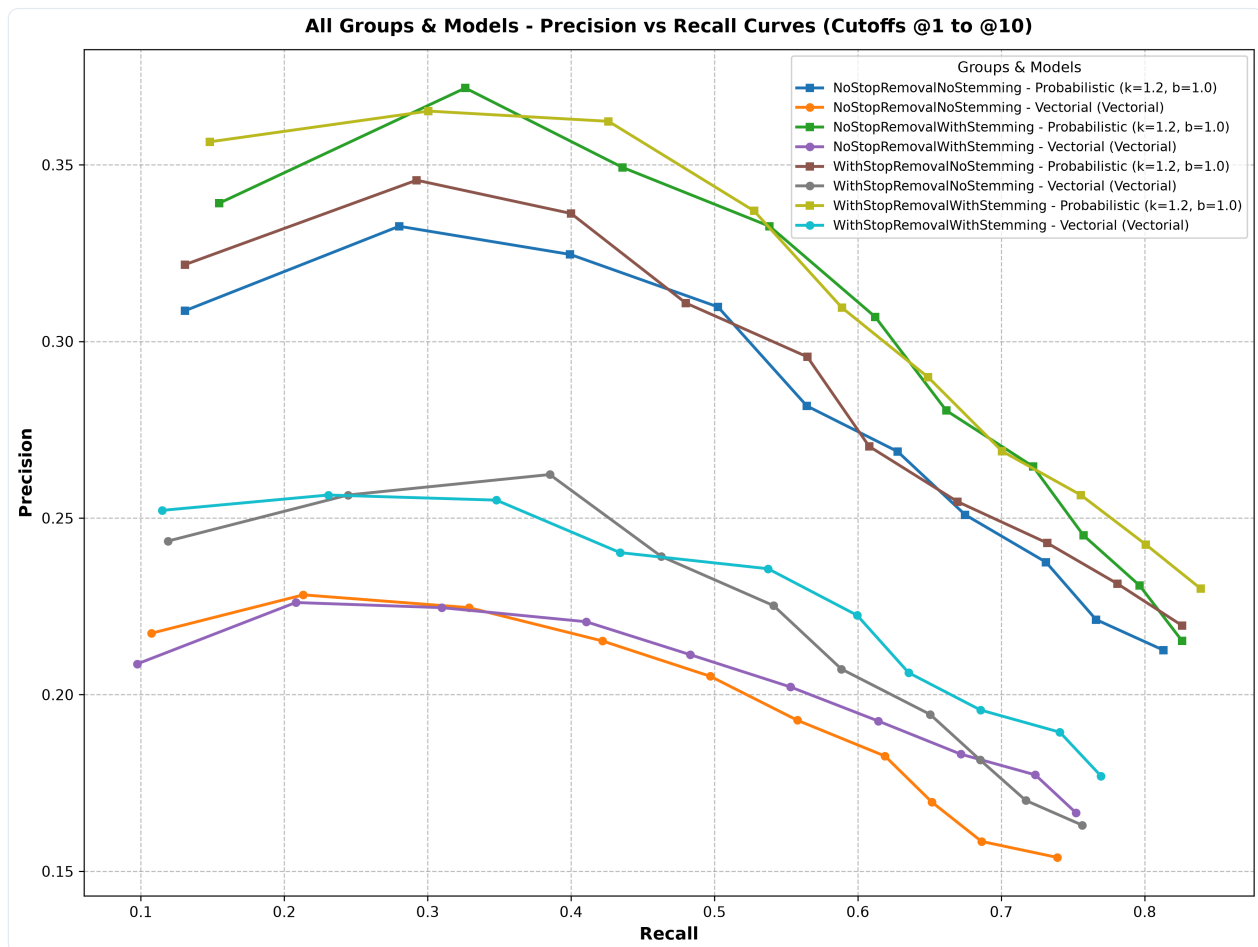
$$b \in \{0, 0.75, 1\}$$

Comparação entre Modelo Vetorial e Modelo Probabilístico

Para realizar a comparação entre os dois modelos, o modelo probabilístico recebeu parâmetros fixos $k = 1.2$ e $b = 1.0$. As métricas utilizadas para comparação foram Precision, Recall e MAP.

Curva Precision x Recall

Na curva do gráfico Precision x Recall, cada ponto representa um top-k, com $P@k$ e $R@k$ definidos como a média de todas as queries executadas.



Analisando a figura, é fácil averiguar que, em todos os casos, **o modelo probabilístico foi superior ao vetorial** com uma boa margem entre o pior do modelo probabilístico e o melhor do modelo vetorial (cerca de 5%).

Para o modelo **probabilístico**, existem duas duplas de grupos semelhantes: `WithStopRemovalWithStemming` e `NoStopRemovalWithStemming` apresentaram desempenho semelhante e claramente superior a `WithStopRemovalNoStemming` e `NoStopRemovalNoStemming` (também semelhantes). Isso evidencia que o tratamento de strings que afeta mais positivamente o modelo é a aplicação do stemming. Ao final da curva, é possível ver que a remoção de stop words forneceu uma pequena vantagem de modo geral.

Para o modelo **vetorial**, também existem duas duplas de grupos semelhantes, mas comportamento inverso: `WithStopRemovalWithStemming` e `NoStopRemovalNoStemming` foram superiores a `WithStopRemovalNoStemming` e `NoStopRemovalWithStemming`. Isso evidencia que, no modelo vetorial, o tratamento mais significativo é a remoção de stop words. O stemming fornece apenas uma pequena vantagem de modo geral.

Pela análise da curva Precision x Recall, pode-se concluir que:

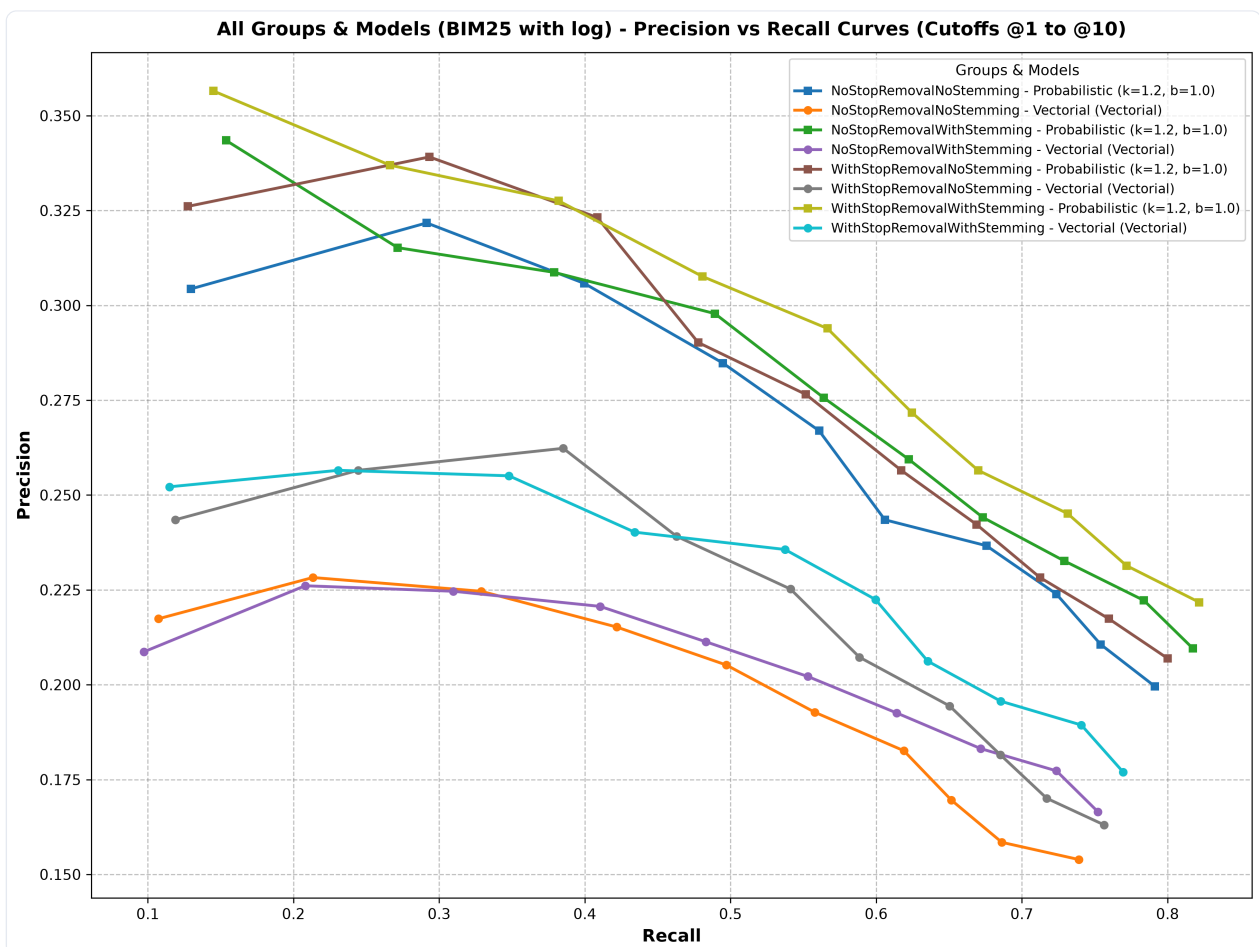
- No modelo **probabilístico**, o tratamento mais relevante é o stemming. A remoção de stop words possui baixa relevância, sendo mais decisivo nos primeiros resultados e se tornando quase imperceptível nos resultados posteriores.
- No modelo **vetorial**, o tratamento mais relevante é a remoção de stop words. O stemming possui baixa relevância, mas ainda assim é um fator discrepante ao longo de toda a curva.

Através dessas afirmações, pode-se inferir a causa.

No modelo **probabilístico**, a variação pequena ao alternar a aplicação da remoção de stop words evidencia que o modelo lida muito bem com palavras que se repetem em quase todos os arquivos, conseguindo reduzir sua relevância ao nível esperado. Isto provavelmente se deve ao parâmetro de penalização por **tamanho do documento**.

Já a grande variação na presença do stemming pode transparecer um defeito: O modelo é facilmente influenciável pela frequência dos termos, ganhando eficiência quando os termos são agrupados (“andar”, “andou”, “andei” -> “and”), o que retira a influência de termos únicos. Para testar essa hipótese, foi aplicado **log** aos termos de frequência da equação do BM25, de modo que:

$$\text{score}(d, q) = \sum_{t \in q} \text{IDF}(t) \frac{\log(1 + tf(t, d))(k_1 + 1)}{(\log(1 + tf(t, d)) + k_1) \left(1 - b + \left(\frac{b(|d|)}{\text{avgdl}}\right)\right)}$$



O desempenho do BM25 diminuiu. Mas note que a divisão em duplas por presença de stemming já praticamente não existe mais: os desempenhos se aproximaram, de modo que apenas stemming e apenas remoção de stop words apresentaram desempenho próximo a partir de k=5. Isso confirma a sensibilidade do BM25 a termos muito frequentes no documento, mas também mostra que essa sensibilidade é necessária para sua eficiência.

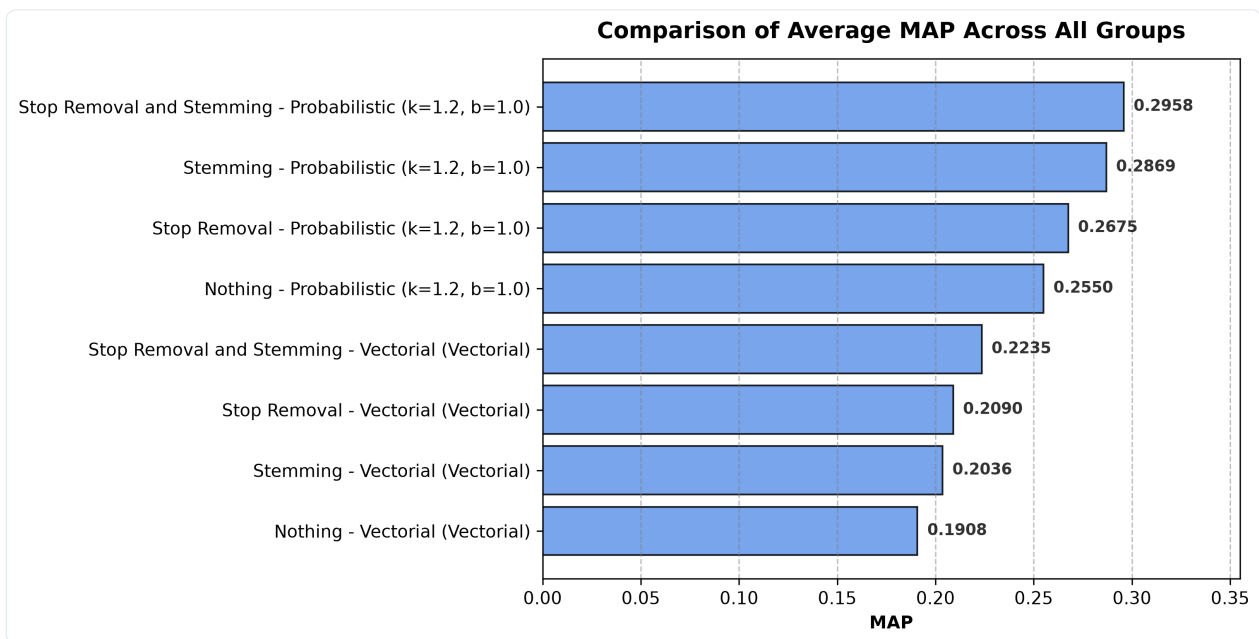
No modelo **vetorial**, a variação pequena ao alternar a aplicação de stemming pode evidenciar um ponto forte do modelo: Quanto a frequência das palavras no documento correlatas à query aumenta muito, o modelo lida bem com isso. Diferente de stop words, palavras com stemming se aproximam e identificam melhor documentos com temas específicos (“medicar”, “medicando”, “medicamento” ->

“medic”). Isso não é tratado pelo **IDF**, mas sim pelo logaritmo que amortece a influência da frequência do termo no peso final.

Já a grande variação ao alternar a remoção de stop words transparece um defeito do modelo vetorial: Como stop words são muito comuns entre query e documento, o vetor de ambos começa a ficar poluído e com ruídos, piorando drasticamente a acurácia do modelo devido a dimensões extras desnecessárias. Isto provavelmente é uma falha da similaridade por cosseno, apesar do termo de controle **IDF**.

Comparação de MAP

Em ordem decrescente, as pontuações MAP para as variações do modelo probabilístico e vetorial se encontram no gráfico abaixo.



Novamente, o modelo probabilístico se mostra superior ao modelo vetorial.

No modelo probabilístico, o stemming se mostrou como mais relevante no incremento do MAP do que a remoção de stop words.

Já no modelo vetorial, a remoção de stop words se mostrou como mais relevante no incremento do MAP do que o stemming, mas com menor discrepância do que no modelo probabilístico.

O comportamento observado na curva de Precision x Recall se repete no MAP, fortalecendo as hipóteses apresentadas.

Análise por consulta

Serão selecionadas 2 consultas para cada modelo onde o resultado foi claramente discrepante (de modo positivo).

Consultas com BM25 superior

Query 52: what is the available information pertaining to the effect of slight rarefaction on boundary layer flows (the ?slip? effect) .

- BM25 MAP: 0.875
- Vetorial MAP: 0.1207943404634581

1. Doc 326 [RELEVANT]

first-order slip effects on the compressible laminar boundary layer over a slender body of revolution in axial flow .
analysis of the compressible boundary layer with transverse curvature in first order slip flow . no boundary-layer interaction effects are considered and only the zero pressure-gradient case is examined .

2. Doc 21 [RELEVANT]

on heat transfer in slip flow .
a number of authors have considered the effect of slip on the heat transfer and skin friction in a laminar boundary layer over a flat plate .
reference 1 considers this by a perturbation on the usual laminar boundary-layer analysis while some other studies.dash e.g., reference the impulsive motion of an infinite plate .

3. Doc 306 [RELEVANT]

approximation to laminar compressible boundary layer on flat plate in slip flow .
the first-order solution for the laminar compressible boundary-layer flow over a flat plate at constant wall temperature is given . the effect of slip at the wall as well as the interaction between the boundary-layer flow and the outer stream flow are taken into consideration . the solution is obtained explicitly in terms of the known zero order, or continuum, solution . no assumptions regarding the prandtl number or viscosity-temperature law need be made . it is found that the first-order solution gives a decrease in heat transfer and, for supersonic flow, an increase in skin friction .
for subsonic flow there is no first-order shear effect . the change in heat transfer is due to slip and the change in friction is due to the interaction of the zero- and first-order velocities at the outer edge of the boundary layer .

4. Doc 265 [IRRELEVANT]

some instabilities arising from the interaction between shock waves and boundary layer .
a brief review is made of the available information concerning the flow fluctuations and instabilities arising from shock-induced separation in the flow over aerofoils and wings . the influence this phenomenon has on the oscillatory behaviour of aerofoils and control surfaces is also briefly discussed .
a more detailed consideration is devoted to a recent investigation at the n.p.l. into the part played by shock-induced separation in the instability of a control surface .

5. Doc 1215 [IRRELEVANT]

the effect of slip particularly for highly cooled walls .

it is found that for boundary conditions on the velocity slip and on the temperature jump which are not oversimplified in an unrealistic way, the effect of these phenomena on the heat transfer and the shear at a stagnation point is of the order of the ratio of the mean free path outside the boundary layer to the boundary-layer thickness, even for highly cooled walls . a simplified theory of this effect is given, which puts the physical reasons for the results in evidence and agrees closely with the more accurate calculations . it is concluded that the effects of slip and jump are not negligible in comparison with other low-reynolds-number corrections, even for very cold walls .

Query 146: does a membrane theory exist by which the behaviour of pressurized membrane cylinders in bending can be predicted .

- BM25 MAP: 0.75
- Vetorial MAP: 0.24285714285714285

1. Doc 1045 [RELEVANT]

the bending strength of pressurized cylinders .

discussion of previously presented experimental data for the loading of pressurized cylinders, in terms of membrane theory .

2. Doc 955 [IRRELEVANT]:

the membrane approach to bending instability of pressurized cylindrical shells .

recent theoretical and experimental research is briefly described to trace the development of deformation and the occurrence of collapse in pressurized circular cylindrical membranes under applied moment loading . the collapse of pure membrane cylinders is then compared with instability of pressurized cylindrical shells . this approach leads to a better understanding of the behavior of pressurized cylinders under bending loads . the results suggest possibilities for further research utilizing the membrane approach .

3. Doc 1130 [IRRELEVANT]:

handbook of structural stability . pt .vi . strength of stiffened curved plates and shells . a comprehensive review of failure

of stiffened curved plates and shells is presented .

panel instability in stiffened curved plates and general instability of stiffened cylinders are discussed . the loadings considered for the plates are axial, shear, and the combination of the two . for the cylinders, bending, external pressure, torsion, transverse shear, and combinations of these loads are considered .

general instability in stiffened cylinders was investigated . for bending and torsion loads, test data and theory were correlated . for external pressure several existing theories were compared . as a result of this investigation a unified theoretical approach to analysis of general instability in stiffened cylinders was developed .

4. Doc 840 [RELEVANT]:

analysis of partly wrinkled membrane .

a theory is derived to predict the stresses and deformations of stretched-membrane structural components for loads under which part of the membrane wrinkles . rather than studying in detail the deformations in the wrinkled region, the present theory studies average displacements of the wrinkled material . specific solutions of problems in flat and curved membranes are presented . the results of these solutions show that membrane structures retain much of their stiffness at loads substantially above the load at which wrinkling first occurs .

5. Doc 854 [IRRELEVANT]

boundary-value problems of the thin-walled circular cylinder .

the homogeneous differential equations of donnell's theory of thin cylindrical shells are integrated . expressions are obtained in closed form for the displacements, membrane stresses, moments, and shear forces .

Consultas com Modelo Vetorial superior

Query 133 : experimental studies of creep buckling .

- BM25 MAP: 0.22897364073834656
- Vetorial MAP: 0.5885462880804495

1. Doc 1019 [RELEVANT]

note on creep buckling of columns .

a method for estimating allowable

load capacities of columns subject to creep is presented . the method, which utilizes approximate stress distributions derived from isochronous-stress-strain curves to estimate column load capacities, is shown to be conservative for the time for which the estimate is made .

an application of the method is made to test data on as-received and on stabilized 24s-t4 aluminum alloy . a comparison of the computed column capacities with experimental capacities indicates that the method is satisfactory for estimating the decrease in capacity with increasing time .

easily obtained, time-dependent tangent-modulus loads are discussed . they are interpreted as being approximations to allowable load-capacity estimates . a limited application is made to test data, and the results appear promising . it is concluded that if certain limitations are recognized, the method may prove to be useful because of its simplicity .

a presentation of the results of an experimental investigation of the effects of column imperfection and column-material variation is made . it is found that column-capacity variations of the order of 10 per cent can result from column-imperfection differences and column-material variation .

the results of an experimental study of the variation of column capacity with temperature of exposure are presented . they indicate that column efficiency, as measured by decrease in capacity, can be acceptable for very short times at the higher temperatures . the efficiency at these higher temperatures falls rapidly, however, with increasing time .

2. Doc 1016 [RELEVANT]

principles of creep buckling weight-strength analysis .

the relation of the time-dependent tangent-modulus load--as conceived by shanley--to actual column capacity is clarified . it may be interpreted as a limiting case of the conservative estimate . the time-dependent tangent-modulus load is, therefore, an approximation to a conservative estimate . the approximation, however, may

be either conservative or nonconservative when applied to imperfect or real columns . typical cases are discussed and experimental results for two alloys are cited .

3. Doc 951 [IRRELEVANT]

a unified theory of creep buckling under normal loads .

a general theory of creep buckling, with the initial imperfection as a parameter, is developed for the case of normal loading . a hyperbolic-sine law is used to describe the process of creep . the theory is believed to be applicable to, among other structures, columns, tubes, and possibly conical shells . the wall of the structure is idealized as a sandwich in order to simplify the integration of the equations .

experimental data on columns and tubes, from two different sources, are compared with the predictions of the theory .

4. Doc 1018 [RELEVANT]

note on creep buckling of columns .

the results of short-time elevated-temperature creep tests of objective of obtaining procedures for predicting column lifetime . semiempirical lifetime curves are obtained with the aid of a previously published column creep theory and are used for deriving column curves . the semiempirical lifetime curves are also used to study the effect of varying applied stress and out-of-straightness . in the range considered, small variations in out-of-straightness are found to be of little practical significance, whereas, small stress variations change the column lifetime considerably . for the range of out-of-straightness encountered in the tests, the data can be presented in plots that do not explicitly include out-of-straightness, and plots of this type should be satisfactory for predicting column lifetime for design purposes .

5. Doc 1025 [IRRELEVANT]

note on creep buckling of columns .

the creep of a slightly crooked section column carrying a constant load is studied theoretically . the material of the column is characterized by a strain-time relationship, under constant uniaxial stress, of the form, where ϵ is the total strain, σ is the constant stress, t is the time, and e, a, b , and k are material constants . this form was selected because it applies to at least two alloys--75s-t6 aluminum alloy at 600 f . and a low-alloy steel at 800 and 1,100 f . however, the analysis is intended for any material having creep properties of the same form and for which the material constants are known . a strain-time relationship under variable uniaxial stress, necessary for the column analysis, is formulated from the constant-stress properties with the aid of shanley's engineering hypotheses of creep .

the analysis leads to the conclusion that the lateral deflection approaches infinity--that is, the column collapses--in finite time .

results are given showing the maximum length of time the column can support a given load before it collapses and the growth of stresses, strains, and deflections prior to collapse .

Query 167 : exact solution methods for calculating the ablative mass loss of a material ablating at high temperatures in a hypersonic flight environment .

- BM25 MAP: 0.41666666666666663
- Vetorial MAP: 0.75

1. Doc 274 [RELEVANT]

analysis of quartz and teflon shields for a particular re-entry mission .

the transient performance of ablation type heat protection shields is treated herein for the surface of a vehicle returning from outer space to the earth . the vehicle weighs 8640 kg, has a ballistic factor of 500 lb ft, re-enters with a speed of 11 km sec at ratio of 0.5, and is subjected to a maximum deceleration of 7.7 times the gravity constant .

by use of well known equations for the heat transfer and the mass transfer at a heated surface, a numerical calculation method is derived which, for the investigated ablation processes, yields exact transient solutions of the fundamental system of partial differential equations . the method is applied to various quartz shields and to one teflon shield, which all evaporate so readily under the conditions of the problem at hand that practically no flow of molten shield material exists . the solutions also show comparatively small temperature changes parallel to the surface .

the results show that the nose of the vehicle is cooled predominantly by the evaporation of the quartz or the teflon,. the rest of the vehicle's surface is cooled by radiation of the quartz or evaporation of the teflon . the large mass transfer effects on the nose of the vehicle are detrimental since the resulting low surface temperatures prevent the radiative heat transfer out of the shield, which does not involve any

mass loss, from being the desirable governing cooling factor .

2. Doc 1279 [IRRELEVANT]

sublimation in a hypersonic environment .

a priori knowledge of the response of materials subjected to a severe aerothermal environment is essential in the space age . the successful design of space and re-entry vehicles demands that the fundamental problem of the interaction between a material and dissociated air be properly formulated and solved . in this paper, the problem of sublimation in a hypersonic environment is considered .

in this study of hypersonic ablation, the pertinent conservation equations are derived and the simultaneous processes of diffusion, convection, and thermal exchange are analyzed for the vaporization of a refractory material which is subjected to the environmental conditions encountered during hypersonic reentry .

for simplicity, only the forward stagnation point of an axially symmetric body is treated . it is shown that the quantity, called the effective heat of vaporization, which includes all heat absorbing or heat blocking effects, is an increasing function of flight speed, independent of body size, except where nonequilibrium vaporization effects or radiative effects appear .

3. Doc 908 [IRRELEVANT]

random vibration .

random vibration is vibration which results from an excitation which is not well represented by any simple function (sinusoid, step, etc.) or any simple combination of such functions but which is satisfactorily modeled by a stochastic process . it is perhaps not too much of an exaggeration to say that /all vibration is random vibration ./ every vibration record contains /hash/ at some level . nevertheless, until recently, engineering vibration theory has been able to get along without including the consideration of random excitations .

now in several fields simultaneously there has occurred a burst of activity in the application of random processes . the response of aircraft to buffeting from atmospheric turbulence and the response of ships to confused seas have been put on reasonably firm footing . possibly the most dramatic problems have been posed by the development of large jet and rocket engines which produce spectacular amounts of random vibrational energy . the high level of random vibration in a jet plane or a missile provides a severe environment with respect to fatigue failure of structural members and with respect to malfunctions of sensitive equipment .

4. Doc 82 [RELEVANT]

theoretical investigation of the ablation of a glass-type heat protection shield of varied material properties at the stagnation point of a re-entering irbm .

the melting-type heat protection at the stagnation point of a re-entering irbm is treated by employing homogeneous, opaque, and nondecomposing glass shields which do not exceed a temperature of some effects due to variations of the glass properties . the ballistic re-entry vehicle has a nose diameter of 0.635 m, a ballistic factor of 3.5×10 , a re-entry angle of 124.9 (from the vertical) at an altitude of 100 km, and a re-entry speed of 4.5 . the performance of 36 different glass shields with assumed combinations of material properties is investigated by employing a calculation method which yields practically exact, transient solutions for the problem . as a corollary, results for a certain steady flight state are also given . the discussions made it possible to derive under realistic flight conditions some thermal characteristics for the employment of thin, or light-weight, glass shields .

investigation of these hypothetical glass shields leads to the conclusion that a low thermal conductivity and a high specific heat, and thus, a small thermal diffusivity are most desirable . a small thermal diffusivity yields high surface temperatures, causing a high radiative heat transfer out of the shield,. and steep temperature profiles normal to the surface, causing a small thermal penetration across the shield with little total ablation of the shield . results show that for the assumed irbm re-entry, the necessary thickness of the employed glass shields increases monotonically with thermal diffusivity which is the only material parameter affecting this thickness .

a high viscosity level and a high emissivity constant of the surface of the supposedly opaque shield are also desirable,. although, these two properties exert a comparatively small influence on the overall performance when disregarding glass shields with an extremely low viscosity level .

5. Doc 576 [IRRELEVANT]

viscous and inviscid stagnation flow in a dissociated hypervelocity free stream .

high reynolds number hypersonic stagnation flow over a blunt-nosed body in a nonequilibrium dissociated free stream is analyzed and compared to a similar flow in an initially undissociated ambient gas . free stream dissociation effects on various equilibrium stagnation flow properties in air are presented as a function of the ambient atom mass fraction and dissociation energy for velocities ranging from 15,000 to 25,000 fps . significant changes in the bow shock geometry, stagnation gas state, and boundary layer behavior are found when the free stream dissociation involves more than 10% of the total energy . it is observed that for large amounts of both atomic oxygen and nitrogen ahead of the body, the equilibrium shock layer properties converge toward those pertaining to chemically and vibrationally-frozen flow across the bow shock .

moreover, under certain conditions, the ionization level can be increased by an order of magnitude and the usual reduction in frozen boundary layer heat transfer due to a highly-cooled noncatalytic surface can increase from stall of adjacent stages .

the effects of compromises of stage matching to favor part-speed operation were also considered . this phase of the study indicated that such compromises would severely reduce the complete-compressor-stall margin . furthermore, the low-speed stage stall problem is transferred from the inlet stages to the middle stages, which are more susceptible to abrupt-stall characteristics .

the analysis indicates that inlet stages having continuous performance characteristics at their stall points are desirable with respect to part-speed compressor performance . these characteristics must, however, be obtained when the stages are operating in the flow environment of the multistage compressor . alleviation of part-speed operational problems may also be obtained by improvement in either stage flow range or stage loading margin .

the results of this analysis are only qualitative . the trends obtained, however, are in agreement with those obtained from experimental studies of high-pressure-ratio multistage axial-flow compressors, and the results are valuable in developing an understanding of the off-design problem . in addition to these stage-matching studies, a general discussion of variable-geometry features such as air bleed and adjustable gas model . numerical solutions of non-equilibrium airflows with fully coupled chemistry provide a preliminary verification of such scaling for benser, w.a.

limit characteristics . the analysis indicated that all these problems could be attributed to discontinuities in the performance characteristics of the front stages . such discontinuities can be due to the type of stage stall or to a deterioration of stage performance resulting blades is included .

Consultas com desempenho insatisfatório em ambos

Query 31 : what size of end plate can be safely used to simulate two-dimensional flow conditions over a bluff cylindrical body of finite aspect ratio .

- BM25 MAP: 0.000741839762611276
- Vetorial MAP: 0.0007668711656441718

1. Doc 751 [IRRELEVANT]:

a note on the use of end plates to prevent three dimensional flow at the ends of bluff cylinders .

the results are given of some observations of the effects of end plates on the three-dimensional separated flow at the ends of cylindrical models . while these are by no means exhaustive, it is felt that they are of sufficient interest to merit putting on record .

2. Doc 1153 [IRRELEVANT]:

a study of the simulation of flow with free stream mach number 1 in a choked wind tunnel .

the degree to which experimental results obtained under choking

conditions in a wind tunnel with solid walls simulate those associated with an unbounded flow with free-stream mach number 1 is investigated for the cases of two-dimensional and axisymmetric flows . it is found that a close resemblance does indeed exist in the vicinity of the body, and that the results obtained in this way are generally at least as accurate as those obtained in a transonic wind tunnel with partly open test section . some of the results indicate, however, that substantial interference effects, particularly those of the wave reflection type, may be encountered under certain conditions, both in choked wind tunnels and in transonic wind tunnels, and that the reduction of these interference effects to acceptable limits may require the use of models of unusually small size .

3. Doc 1209 [IRRELEVANT]:

aerodynamic processes in the downwash-impingement problem .

theoretical and experimental data relating to the downwash impingement problem are examined in order to arrive at a coherent understanding of the process of entrainment of ground particles in the flow . it is demonstrated that a key mechanism in the process is the interaction of nonuniform flow in the ground boundary layer with bluff ground particles . this interaction produces a lift force which, under typical conditions, equals or exceeds the particle weight .

in the interest of quantitative prediction of the conditions necessary for particle entrainment, four subsidiary problem areas in the impinging jet are examined . these are the viscous decay, the inviscid flow field, the ground boundary layer, and the forces on a bluff body in nonuniform flow . applicable theories are used in conjunction with experimental data to assess the accuracy and range of validity of the theories, and to define the stream conditions which will cause particle entrainment .

available data are applied to the establishment of criteria for particle entrainment in the vicinity of the impinging-jet stagnation point . these criteria show that entrainment occurs in a finite annular region on the ground plane, and that the particles most readily entrained are those with a diameter equal to about two-thirds the thickness of the ground boundary layer . the configuration size is shown to influence the process in that the onset of entrainment is fixed by the jet diameter and velocity, and the size of the ground particles . the criteria established provide a quantitative estimate of the conditions causing entrainment and provide a basis for scaling experimental results to a variety of full-scale situations .

4. Doc 228 [IRRELEVANT]

navier-stokes solutions at large distances from a finite body . this paper is concerned with a theoretical investigation of the flow field at large distances from an object moving through a viscous fluid . the discussion will be restricted to the case of two-dimensional stationary incompressible flow . the object will be assumed to be of finite size . the domain of the fluid is infinite and it is assumed

that there are no other boundaries for the fluid except that of the given object . the reynolds number will be assumed to have a fixed value., thus we shall not consider the limiting cases of the reynolds number tending to zero or to infinity .

5. Doc 1245 [IRRELEVANT]

some aspects of nonequilibrium flows .

in this paper are discussed some of the general features of nonequilibrium flow . in particular, vibrational relaxation is discussed in detail . this case is somewhat simpler than dissociation and ionization but it illustrates some of the main new features of nonequilibrium flow . those aspects of two-dimensional and axisymmetric flow behind shock waves are examined analytically which yield significant information without requiring numerical solution of the governing equations .

the thermodynamics of a vibrational relaxing gas are discussed . the conditions for simulating flows are noted . crocco's theorem and the characteristic equations are derived . then a simple method of obtaining the initial gradients of the flow variables behind a shock is shown . these gradients are used in discussing two particular flows . an exact solution for flow over a cusped body is obtained . flow over a wedge near the tip and far from the tip is considered . it is found that far from the tip a boundary-layer type phenomenon occurs .

Query 22 : did anyone else discover that the turbulent skin friction is not over sensitive to the nature of the variation of the viscosity with temperature .

- BM25 MAP: 0.0010214504596527069
- Vetorial MAP: 0.0017094017094017094

1. Doc 125 [IRRELEVANT]

measurements of skin friction of the compressible turbulent boundary layer on a cone with foreign gas injection .

measurements of average skin friction of the turbulent boundary layer have been made on a 15 total included angle cone with foreign gas injection . measurements of total skin-friction drag were obtained at free-stream mach numbers of 0.3, 0.7, 3.5, and $\times 10$ with injection of helium, air, and freon-12 through the porous wall . substantial reductions in skin friction are realized with gas injection within the range of mach numbers of this test . the relative reduction in skin friction is in accordance with theory--that is, the light gases are most effective when compared on a mass flow basis . there is a marked effect of mach number on the reduction of average skin friction,. this effect is not shown by the available theories . limited transition location measurements indicate that the boundary layer does not fully trip with gas injection but that the transition point

approaches a forward limit with increasing injection . the variation of the skin-friction coefficient, for the lower injection rates with natural transition, is dependent on the flow reynolds number and type of injected gas,. and at the high injection rates the skin friction is in fair agreement with the turbulent boundary-layer results .

2. Doc 560 [IRRELEVANT]

a theoretical study of the effect of upstream transpiration-cooling on the heat transfer and skin friction characteristics of a compressible laminar boundary layer .

an analysis is presented which predicts the skin-friction and heat-transfer characteristics of a compressible, laminar boundary layer on a solid flat plate preceded by a porous section that is transpiration cooled . the analysis is restricted to a prandtl number of unity and linear variation of viscosity with temperature .

the local skin friction has been found to have a low value in the region of transpiration cooling and then to increase until it approaches the value for a completely nonporous surface asymptotically . the initial increase in local skin friction is rapid as half of the ultimate increase occurs in a distance beyond the porous region that is about 20 percent of the length of the porous region for all rates of injection . when the total coolant flow rate is kept constant and the porous length is varied, it is found that the average skin friction on a partially porous plate is slightly lower than that on a fully porous plate .

the local heat transfer behaves in a manner similar to that of the local skin friction . it is found, in an example, that the temperature at the end of a partially porous plate could be maintained at about the same temperature as a fully porous plate by doubling the total rate of coolant flow .

3. Doc 254 [IRRELEVANT]

boundary layers with suction and injection . a review of published work on skin friction .

available data on the effects of suction and injection on skin

friction are summarised and compared .

it is shown that injection into a turbulent boundary layer can produce a skin friction coefficient lower than the laminar value at the same reynolds number on an impermeable plate .

4. Doc 413 [IRRELEVANT]

turbulent skin friction at high mach numbers and reynolds numbers in air and helium . nasa r82, 1960 .

results are given of local skin-friction measurements in turbulent boundary layers over an equivalent air mach number range from 0.2 to 9.9 and an over-all reynolds number variation of 2×10^5 to 100×10^5 . direct force measurements were made by means of a floating element . flows were two-dimensional over a smooth flat surface with essentially zero pressure gradient and with adiabatic conditions at the wall . air and helium were used as working fluids . an equivalence parameter for comparing boundary layers in different working fluids is derived and the experimental verification of the parameter is demonstrated . experimental results are compared with the results obtained by several methods of calculating skin friction in the turbulent boundary layer .

5. Doc 307 [IRRELEVANT]

an approximate solution of hypersonic laminar boundary layer equations and its application .

approximate formulae of the displacement thickness and the skin friction of the hypersonic laminar boundary layer are derived by use of von karman's integral method, assuming the heat-insulated wall, the prandtl number of unity and chapman and rubesin's formula for the variation of viscosity with temperature .

the results obtained are compared with some exact solutions . because of the good agreement, it seems that these formulae are very useful .

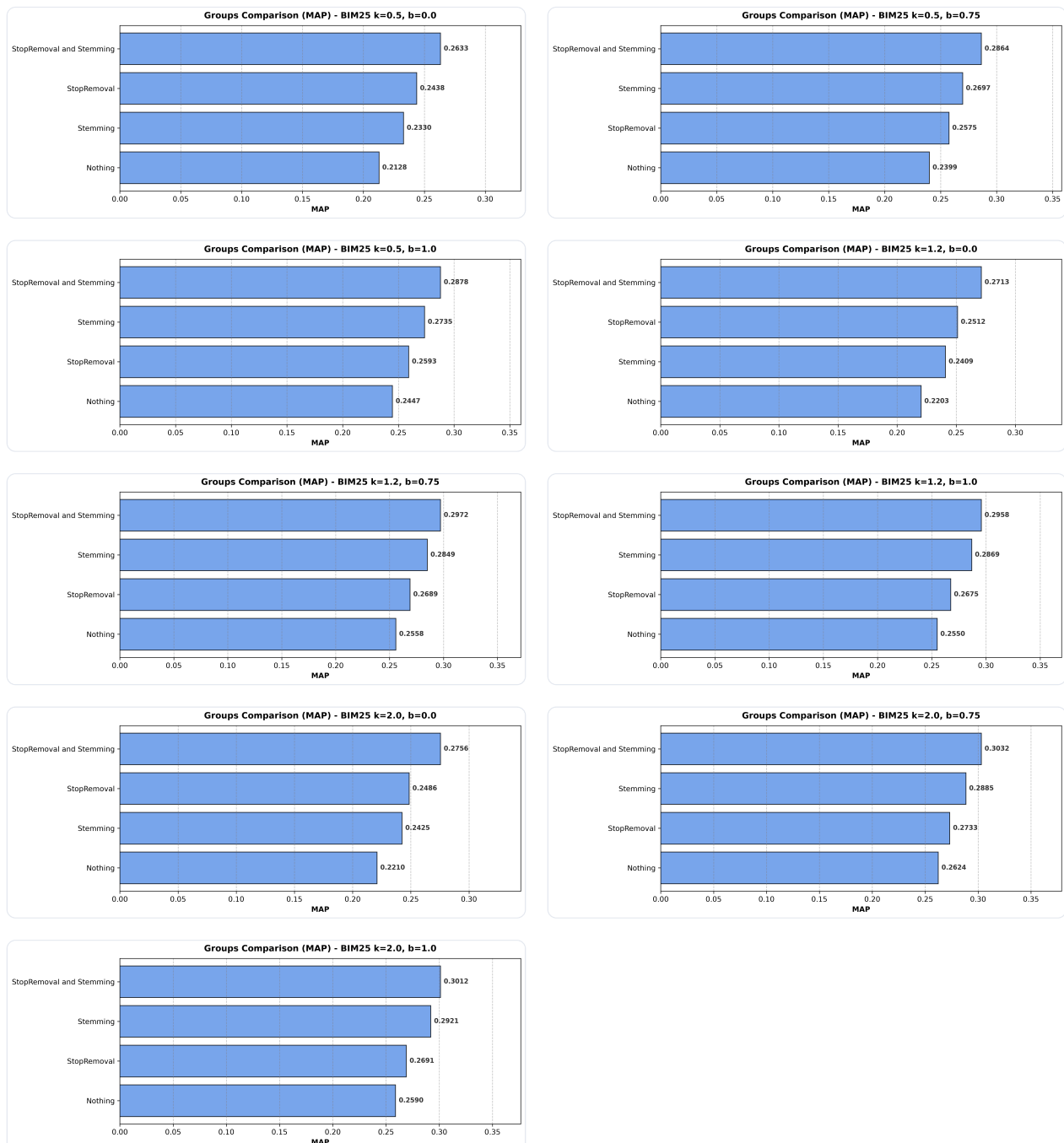
these formulae, together with the tangent-wedge-approximation, are applied to the viscous flow over slender bodies with a sufficiently sharp leading edge . as an example, the pressure distribution over a flat plate is calculated numerically over the entire region of the surface . comparison with other author's theoretical results as well as experimental values is made .

Variação de parâmetros do BM25

Foram estudadas variações dos parâmetros k_1 e b do modelo probabilístico. As variações são concernentes a $k_1 \in \{0.5, 1.2, 2.0\}$ e $b \in \{0, 0.75, 1\}$

MAP agrupado por parâmetros

Primeiro, é útil verificar como cada parâmetro se saiu nos pipelines.



De modo geral, são feitas as seguintes observações (aplicáveis somente ao universo de k e b testados):

1. A remoção de stop words junto com o stemming se estabelece em definitivo como o melhor tipo de processamento para todas as variações do BM25.
2. A ausência da remoção de stop words e de stemming se estabelece em definitivo como o pior tipo de processamento para todas as variações do BM25.
3. Somente stemming foi o segundo melhor processamento em definitivo, exceto quando $b = 0.0$. Nesse caso, a remoção de stop words se define como o segundo melhor processamento.
4. Para todo k , todos os scores são significativamente melhores para $b = 0.75$ em comparação a $b = 0.0$

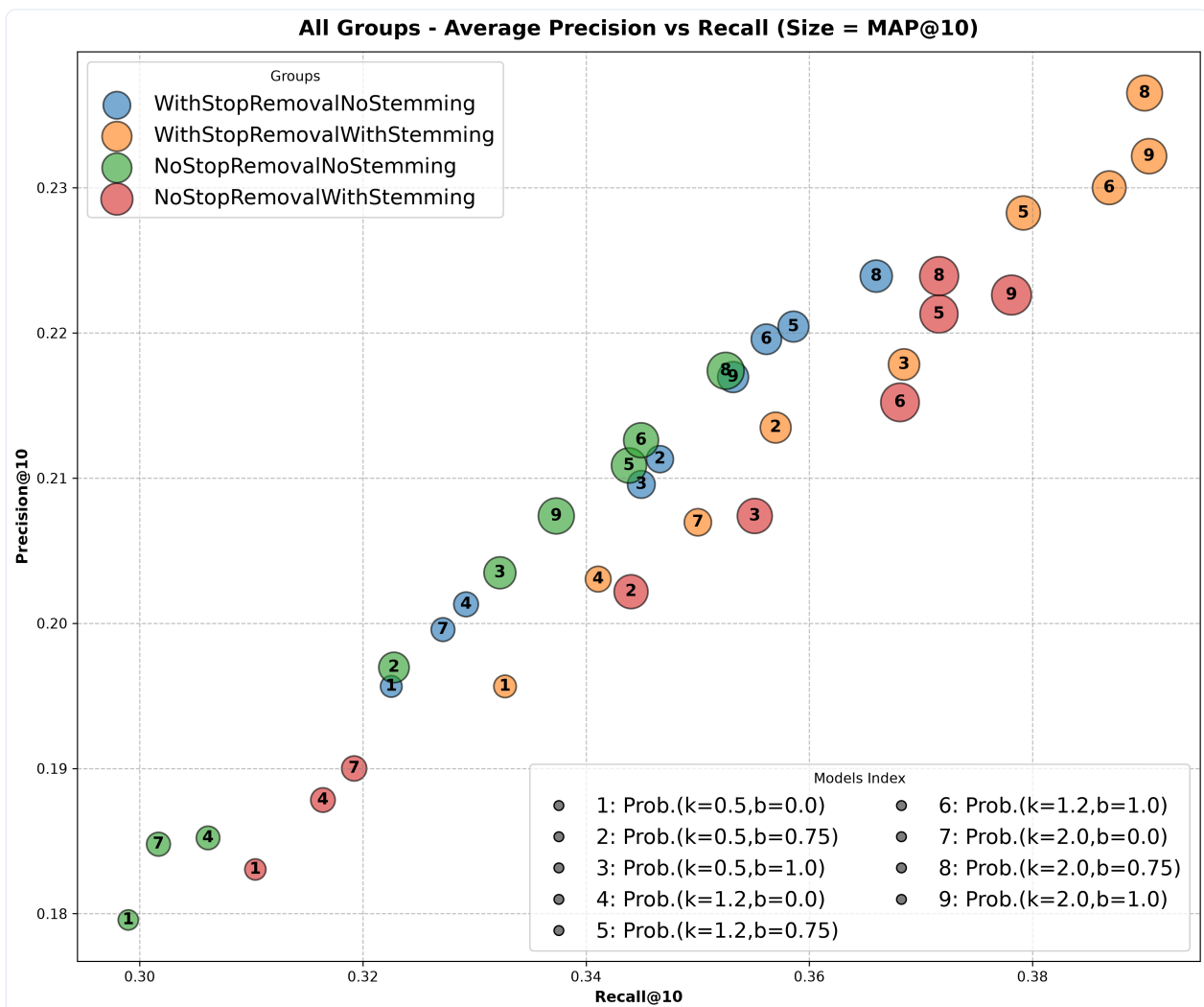
5. Para todo k , todos os scores são desprezivelmente melhores e, por vezes, desprezivelmente piores para $b = 1.0$ em comparação a $b = 0.75$, mas com $b = 0.75$ sempre melhor que $b = 0.0$ para o mesmo k .
6. Para todo b , todos os scores são melhores para $k = 1.2$ em comparação a $k = 0.5$
7. Para $b = 0.75$ e $b = 1.0$, todos os scores são levemente melhores para $k = 2.0$ em comparação a $k = 1.2$. Para $b = 0.0$, isso varia entre leve melhora/piora.

As observações 1 e 2 já eram esperadas. A observação 3 mostra que a ausência do parâmetro b (penalidade por tamanho do documento) representa um problema quando não há remoção de stop words, de modo que somente a remoção de stop words se destaca frente ao stemming. Pela observação 4, também pode-se afirmar que a presença do parâmetro b sempre resulta em melhora significativa na precisão. Por último, a observação 5 revela uma saturação do parâmetro b , de modo que penalizar demais pelo tamanho do documento resulta em perda de precisão.

As observações 6 e 7 revelam, igualmente, uma saturação do parâmetro k .

Mapa de bolhas P@10 x R@10

Ao verificar o comportamento para Precision x Recall ao longo de $k=1$ até $k=10$, a ordem do score das variações se manteve estável, de modo que pode-se analisar somente $k=10$ e inferir o mesmo para $k < 10$.



Em todos os casos, percebe-se, referente ao pipeline, uma ordem de eficiência:

```
WithStopRemovalWithStemming > NoStopRemovalWithStemming > WithStopRemovalNoStemming > .  
NoStopRemovalNoStemming
```

Essa observação foi feita em todas as análises anteriores.

No geral, percebe-se que as variações 5,6,8,9 sempre estão à frente das outras em todos os casos, alternando entre si. Assim, é válido considerá-los as melhores opções possíveis.

O melhor caso possível é `WithStopRemovalWithStemming` com $k = 1.0$ e $b = 0.75$.